

Computer Networks and Data Communication

Module 1

CS 212

Prof. (Dr.) SHAKTI KUNDU

Jan-May 2026

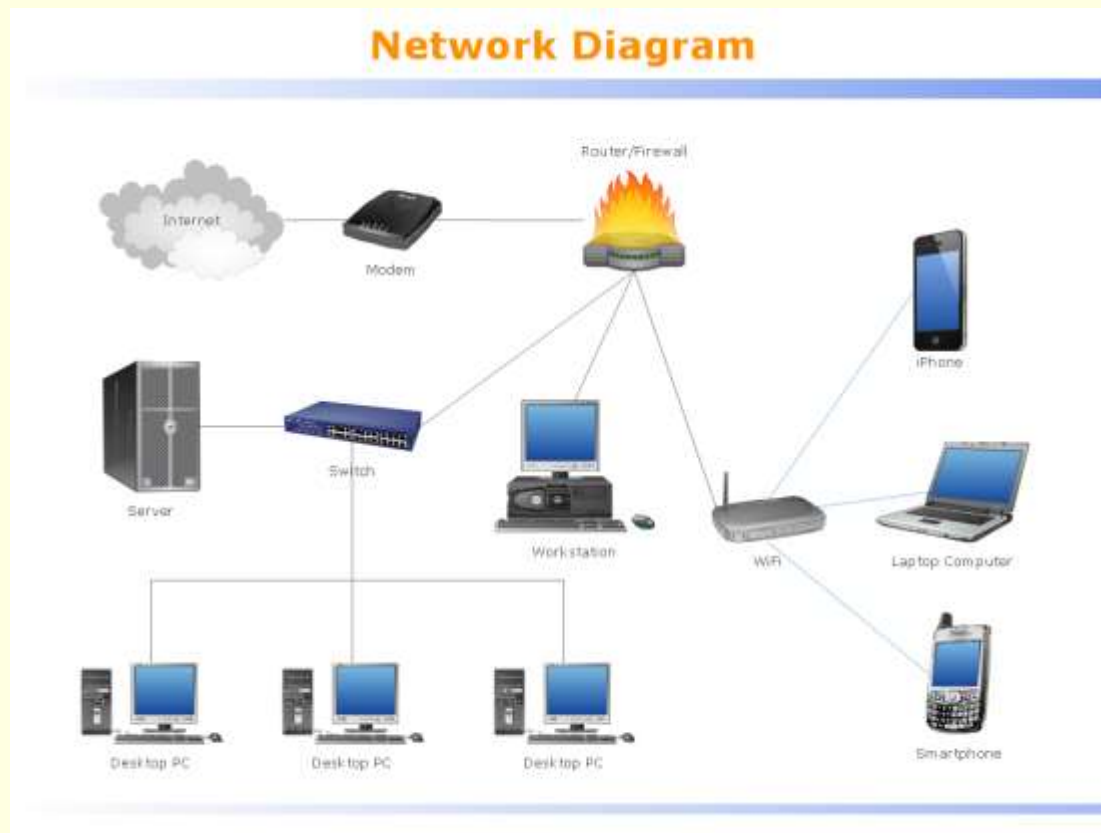
Module wise CONTENTS:

- **Module 1 - Introduction: An overview of networks, the Internet, services and protocols, RFCs and standards of Networking,** Data communication & its components, types of data communication and communication channels. Key components of computer network; network devices; types of computer networks; network topology; reference models (Network Architectures) – OSI & TCP/IP.
- **Module 2 - Physical Layer:** Introduction and functions; types of transmission media – guided and unguided; integrated services digital network (ISDN); asynchronous transfer mode (ATM).
- **Module 3 - Data Link Layer:** Overview; sliding window protocols - Stop-&-Wait; Go-Back-N, Selective Repeat; Error detection and correction; Hamming Code, LRC, VRC, CRC, Checksum, Round Trip Time, Latency (Delay); Medium Access Sublayer: medium access control (MAC) Addresses; Switching techniques - circuit, message & packet; Random Access Protocols - Pure ALOHA, Slotted ALOHA, CSMA / CD, CSMA / CA; Controlled Access Protocols – FDMA, TDMA, CDMA; Channelization Protocols – Reservation, Polling, Token Passing; Ethernet (802.3 IEEE standard), Wireless Links, Wireless LAN - WiFi (802.11).
- **Module 4 - Network Layer:** Introduction; address resolution protocol (ARP); internet protocol (IP); internet control message protocol (ICMP); internet group management protocol (IGMP); IPv4 addressing, representation of IPv6; IPv4 vs IPv6; Conversion IPv4 to IPv6; Routing Protocols - routing information protocol (RIP), open shortest path first (OSPF); border gateway protocol (BGP); Enhanced Interior Gateway Routing Protocol (EIGRP).
- **Module 5 - Transport Layer:** Introduction; transport layer protocols - UDP & TCP, TCP connection establishment; TCP connection termination; TCP congestion control. **Application Layer:** Introduction – features & functions; application layer protocols; Web and HTTP, Email, P2P network, its types & applications.
- **Module 6 – Network Optimization, Mathematical Models, Introduction Advance Computer Networks, MANET, WSN, PCN, SDN, IoT.**

-
- **Module 1 - Introduction: An overview of networks, the Internet, services and protocols, RFCs and standards of Networking,** Data communication & its components, types of data communication and communication channels. Key components of computer network; network devices; types of computer networks; network topology; reference models (Network Architectures) – OSI & TCP/IP.

An overview of computer networks

- A **computer network** is a collection of interconnected devices (computers, servers, routers, switches, smartphones, etc.) that communicate and exchange data using a set of rules called **protocols**.

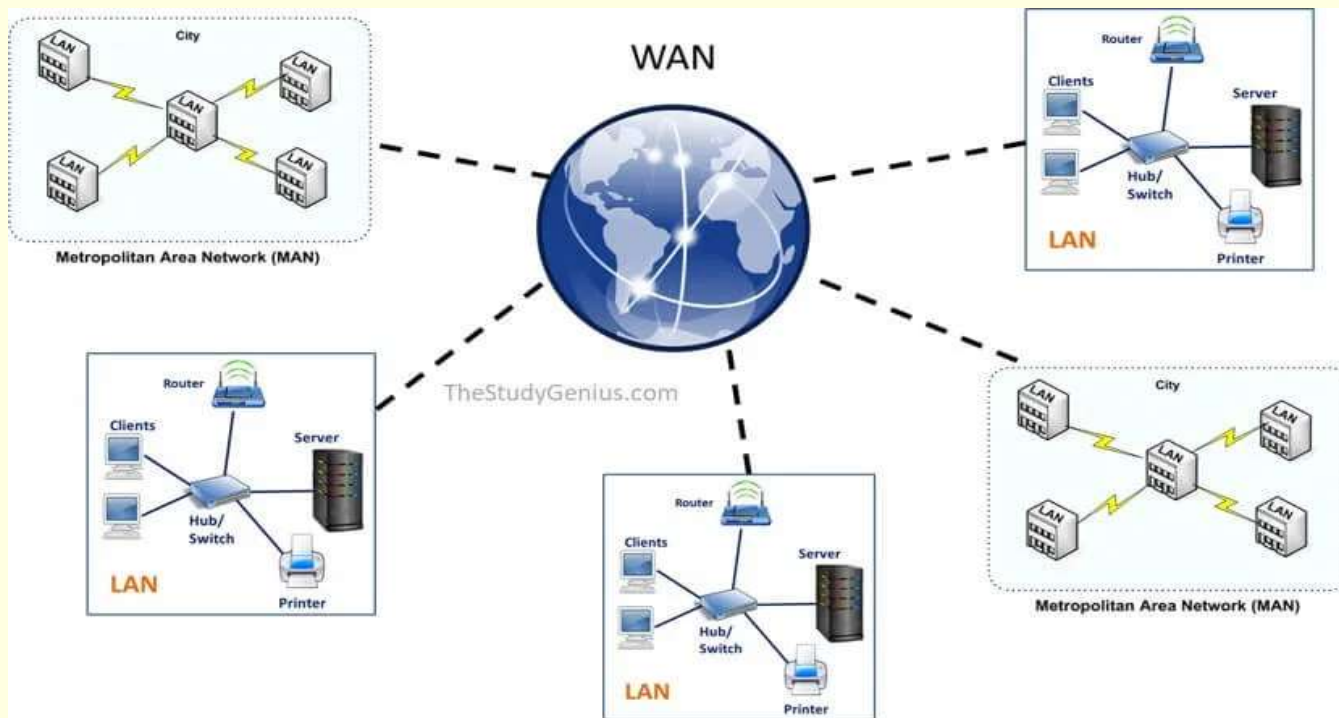


An overview of computer networks

Types of Networks

a. By Geographical Size

- **LAN (Local Area Network):**
Covers a small area—home, office, school.
- **WAN (Wide Area Network):**
Covers large areas—countries or continents (e.g., the Internet).
- **MAN (Metropolitan Area Network):**
Spans a city or campus.



An overview of computer networks

Types of Networks

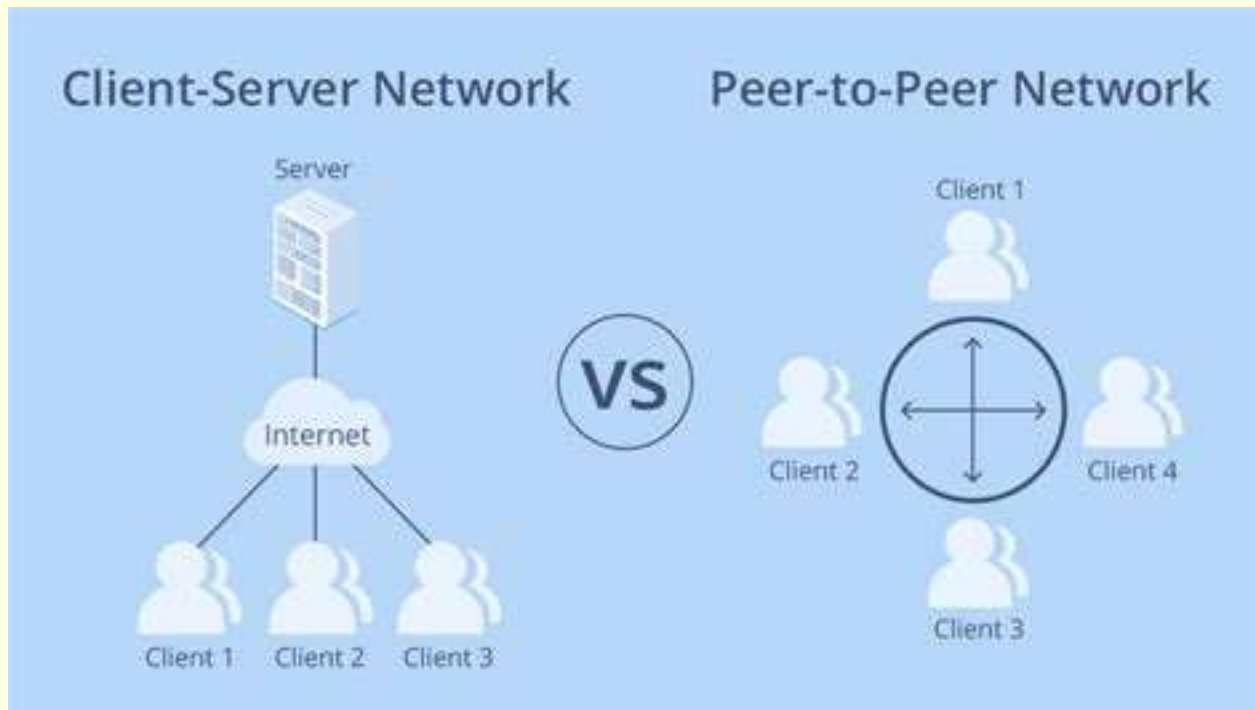
b. By Architecture

- **Client–Server:**

Central server provides services; clients request them.

- **Peer-to-Peer (P2P):**

Each device acts as both client and server.



An overview of computer networks

Key Networking Devices

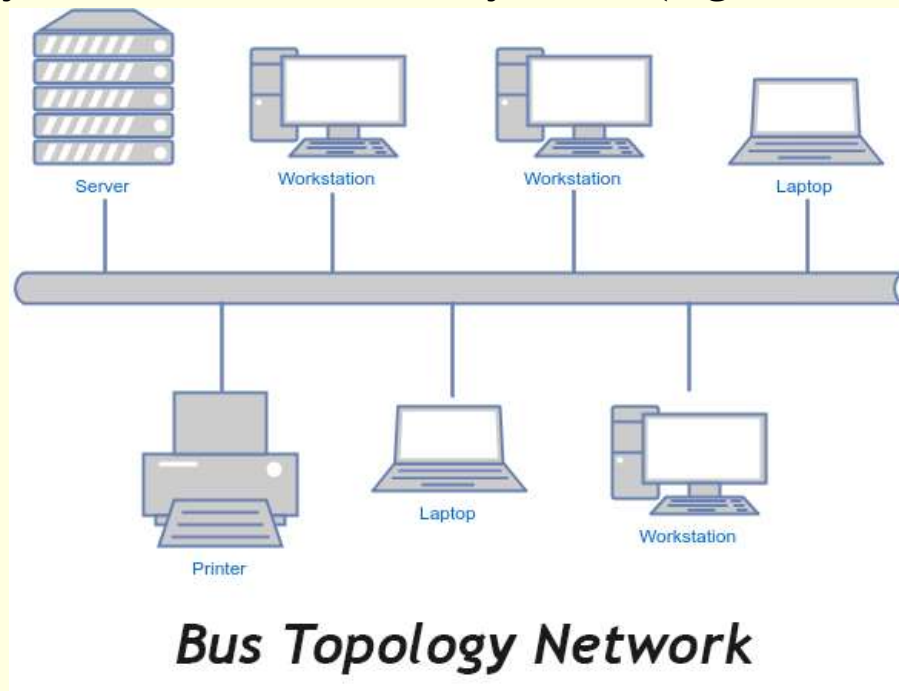
- **Router:** Connects networks and directs traffic.
- **Switch:** Connects devices within a LAN.
- **Access Point:** Provides wireless connectivity.
- **Modem:** Converts signals for Internet access.



An overview of computer networks

Network Topologies

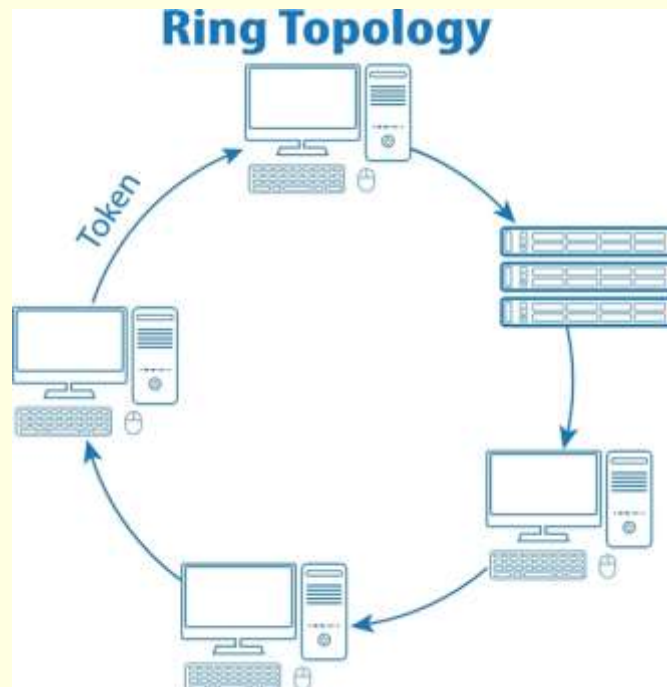
- The physical or logical arrangement of devices:
 - **Bus** – single communication line
 - **Ring** – devices in a circular layout
 - **Star** – all devices connected to a central switch (most common)
 - **Mesh** – every device connects to many others (high reliability)



An overview of computer networks

Network Topologies

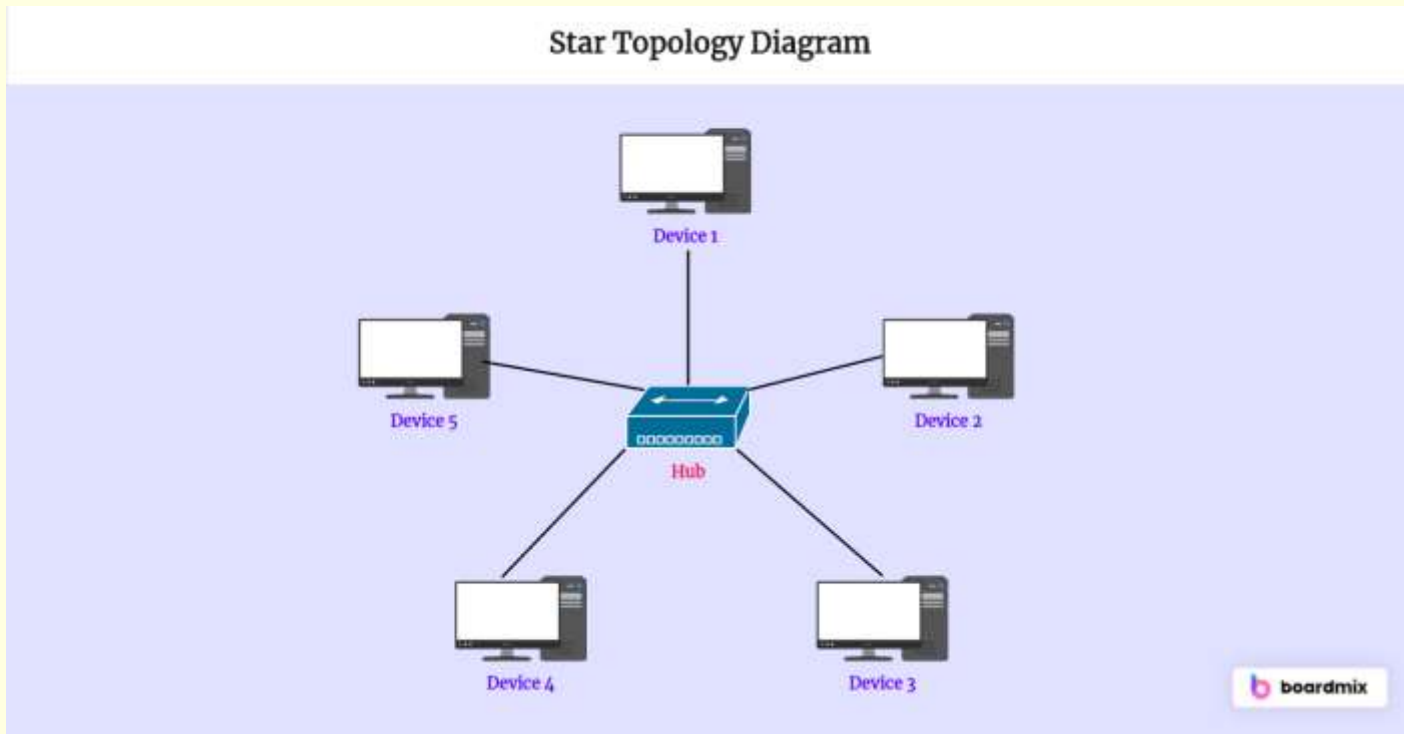
- The physical or logical arrangement of devices:
 - **Bus** – single communication line
 - **Ring** – devices in a circular layout
 - **Star** – all devices connected to a central switch (most common)
 - **Mesh** – every device connects to many others (high reliability)



An overview of computer networks

Network Topologies

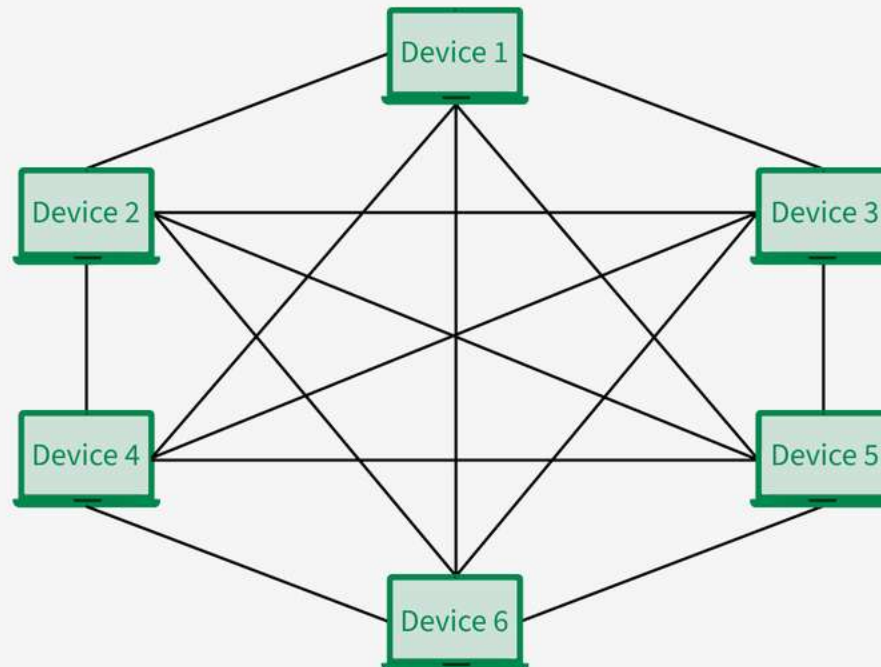
- The physical or logical arrangement of devices:
 - **Bus** – single communication line
 - **Ring** – devices in a circular layout
 - **Star** – all devices connected to a central switch (most common)
 - **Mesh** – every device connects to many others (high reliability)



An overview of computer networks

Network Topologies

- The physical or logical arrangement of devices:
 - **Bus** – single communication line
 - **Ring** – devices in a circular layout
 - **Star** – all devices connected to a central switch (most common)
 - **Mesh** – every device connects to many others (high reliability)

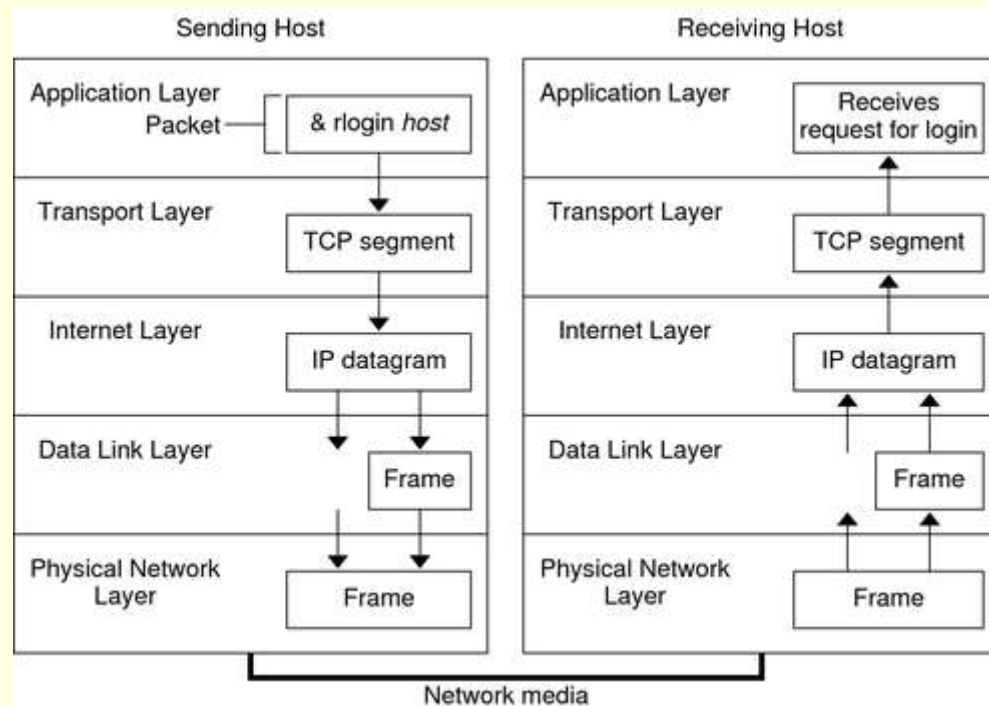


An overview of computer networks

Communication Protocols - Protocols define how data is transmitted.

■ TCP/IP Model Layers

- **Application Layer** – HTTP, FTP, DNS
- **Transport Layer** – TCP (reliable), UDP (fast)
- **Internet Layer** – IP addressing, routing
- **Network Access Layer** – Ethernet, Wi-Fi

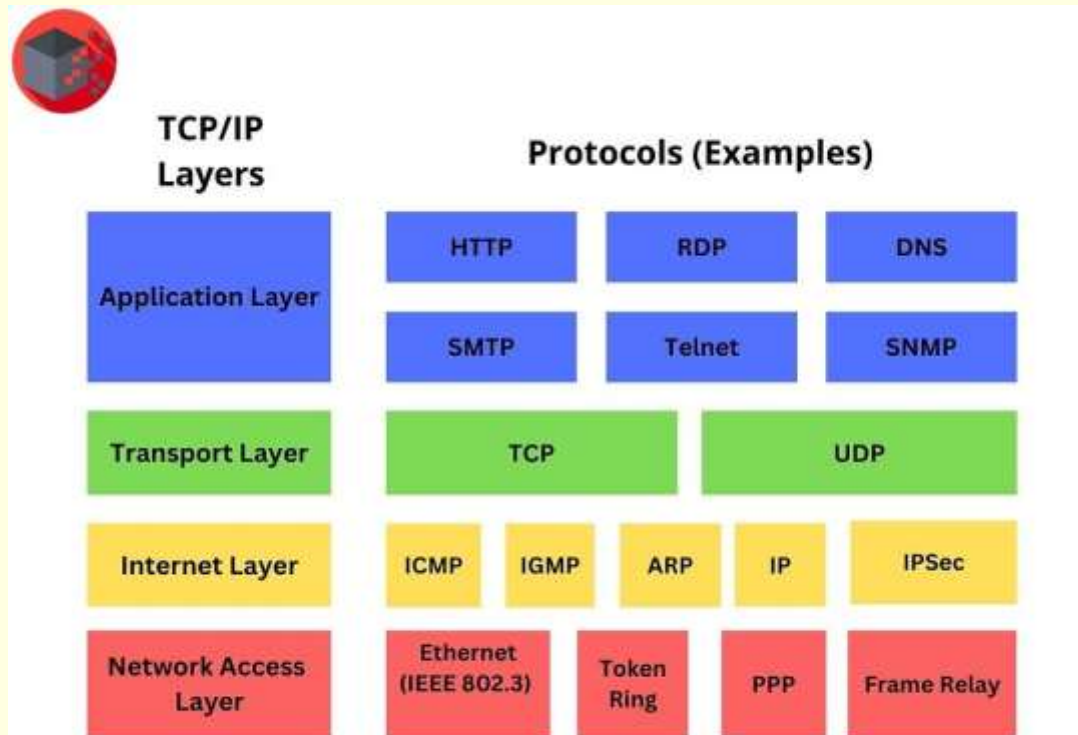


An overview of computer networks

Communication Protocols - Protocols define how data is transmitted.

■ TCP/IP Model Layers

- **Application Layer** – HTTP, FTP, DNS
- **Transport Layer** – TCP (reliable), UDP (fast)
- **Internet Layer** – IP addressing, routing
- **Network Access Layer** – Ethernet, Wi-Fi

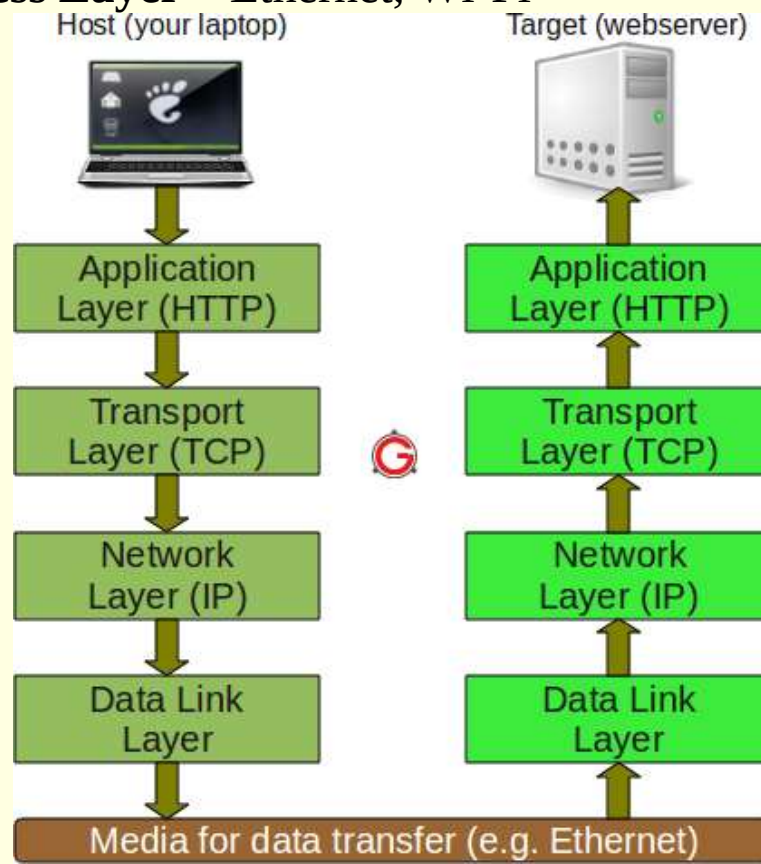


An overview of computer networks

Communication Protocols - Protocols define how data is transmitted.

■ TCP/IP Model Layers

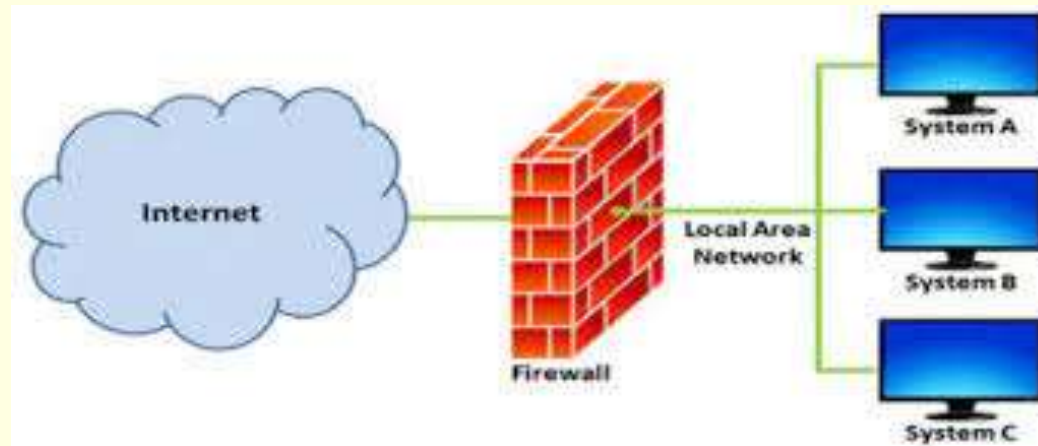
- **Application Layer** – HTTP, FTP, DNS
- **Transport Layer** – TCP (reliable), UDP (fast)
- **Internet Layer** – IP addressing, routing
- **Network Access Layer** – Ethernet, Wi-Fi



An overview of computer networks

Network Security

- Essential for protecting data from cyber threats.
- Common mechanisms:
 - Firewalls
 - Encryption (e.g., HTTPS)
 - Authentication systems
 - VPNs



An overview of computer networks

Wireless vs. Wired Networks

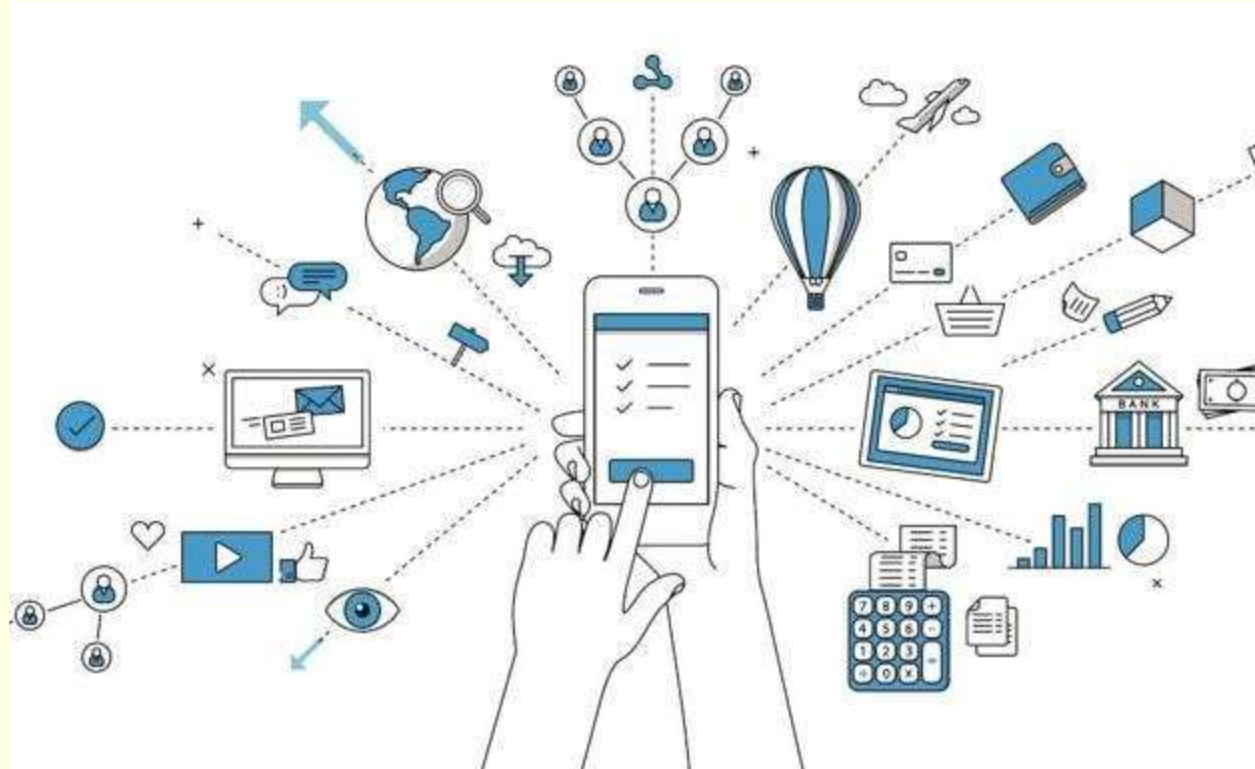
- **Wired (Ethernet):** Faster, more secure, stable
- **Wireless (Wi-Fi):** Convenient, flexible, but more prone to interference and security risks

The Internet

- A global network of networks using TCP/IP.
It supports services like:
- Web browsing
- Email
- Cloud platforms
- Streaming
- Online gaming

the Internet

- The **Internet** is a global system of interconnected computer networks. It allows millions of private, public, academic, business, and government networks to communicate with each other, enabling information to be shared instantly across the world.



the Internet

How the Internet Works

1. Networks Connected Together

- Individual networks (homes, schools, companies) connect to Internet Service Providers (ISPs).
ISPs connect to larger networks, forming a massive worldwide web of connections.

2. IP Addresses

- Every device on the Internet has an **IP address**, which acts like its digital home address.
This allows data to know where it needs to go.

3. Packets

- Information sent over the Internet is broken into tiny pieces called **data packets**.
Each packet may travel a different route but is reassembled at the destination.

4. Routers

- Routers direct packets across networks, choosing the best path for fast delivery.

the Internet

Key Internet Services

- **World Wide Web (WWW):** Websites and web pages accessed through browsers.
- **Email:** Sending and receiving digital messages.
- **Streaming:** Watching videos or listening to music online.
- **Social Networks:** Platforms for communication and sharing content.
- **Cloud Computing:** Storing and processing data on remote servers.
- **Online Gaming:** Real-time multiplayer interactions.

Why the Internet Is Important

- Enables global communication
- Supports business, research, and education
- Provides instant access to information
- Connects devices in the “Internet of Things”
- Powers modern apps and services

Services and Protocols

Common Internet Services and Their Protocols

1. World Wide Web (WWW) Service: Accessing web pages, images, videos, and online applications

■ **Protocols:**

- **HTTP (Hypertext Transfer Protocol):** Basic communication between browser and web server
- **HTTPS:** Secure version of HTTP using encryption

2. Email Service: Sending and receiving electronic messages

■ **Protocols:**

- **SMTP (Simple Mail Transfer Protocol):** Sends email
- **POP3 (Post Office Protocol v3):** Downloads mail from server to device
- **IMAP (Internet Message Access Protocol):** Lets you view and manage mail while stored on the server

Services and Protocols

Common Internet Services and Their Protocols

3. Remote Access Service: Controlling a device from another location

■ Protocols:

- **SSH (Secure Shell):** Secure remote login
- **Telnet:** Older, insecure remote access protocol
- **RDP (Remote Desktop Protocol):** Graphical remote connection (Windows)

4. File Transfer Service: Uploading and downloading files

■ Protocols:

- **FTP (File Transfer Protocol):** Transfers files between systems
- **SFTP/FTPS:** Secure versions of FTP
- **TFTP (Trivial File Transfer Protocol):** Simplified, often used for network booting

Services and Protocols

Common Internet Services and Their Protocols

5. Domain Name System (DNS) Service: Converts human-friendly names (e.g., google.com) into IP addresses

- **Protocol:**

- **DNS:** Uses UDP/TCP to query and respond with IP information

6. Streaming and Real-Time Communication Service: Live audio, video, or gaming

- **Protocols:**

- **RTP (Real-Time Transport Protocol):** Delivers audio/video
- **RTSP (Real-Time Streaming Protocol):** Controls stream playback
- **WebRTC:** Real-time peer-to-peer communication (video calls)

RFCs

- **RFCs (Requests for Comments)** are official documents that describe the rules, standards, procedures, and technologies that define how the Internet works. They are published by the **Internet Engineering Task Force (IETF)** and related groups.
- Although the name “*Request for Comments*” sounds informal, RFCs are actually the authoritative standards that govern Internet technologies such as TCP/IP, HTTP, DNS, email protocols, and many more.



RFCs

What RFCs Are Used For

- RFCs serve several important purposes

1. Internet Standards

- Many foundational Internet protocols are defined in RFCs, such as:
 - **RFC 791 – IPv4**
 - **RFC 2460 / 8200 – IPv6**
 - **RFC 761 / 793 – TCP**
 - **RFC 2616 / 7230 – HTTP/1.1**
- These documents specify how protocols should behave to ensure compatibility across devices and networks.

RFCs

2. Protocol Design

- When engineers create new networking protocols or update existing ones, they write an RFC that describes:
 - How the protocol works
 - The format of messages
 - Expected behavior
 - Security considerations

3. Documentation and Guidance

- Some RFCs provide best-practice guidelines rather than formal standards. For example:
 - Email formatting rules
 - Network security recommendations
 - Routing architecture

RFCs

Types of RFCs

- RFCs are categorized into different types:

1. Standards Track

- These may become Internet Standards.

Levels include:

- **Proposed Standard**
- **Draft Standard**
- **Internet Standard (STD)**

2. Informational

- Provides general information, guidance, or explanations.

3. Experimental

- Describes new ideas being tested.

4. Historic

- Outdated or superseded RFCs.

RFCs

■ How RFCs are Created

- An idea is proposed within IETF working groups.
- Engineers write drafts and revise them.
- The document is reviewed by experts.
- When agreed upon, it is published as an RFC.
- RFCs never change once released—newer RFCs simply *update* or *obsolete* older ones.

■ Why RFCs Matter

- They ensure **interoperability** among Internet devices.
- They provide **transparency** in how the Internet functions.
- They create a **shared reference** for developers, researchers, and network engineers.
- They preserve the **historical evolution** of internet technologies.

Standards of Networking

- Networking standards are **agreed-upon rules and specifications** that ensure computers, devices, and network technologies can work together reliably. Without standards, devices from different companies would not be able to communicate, and the Internet as we know it would not exist.
- Standards define how data is transmitted, how hardware should operate, and how communication protocols function.

Why Networking Standards are Important

- **Interoperability:** Devices from different manufacturers work together
- **Compatibility:** Old and new technologies can coexist
- **Reliability:** Networks function predictably
- **Scalability:** Networks can grow without breaking
- **Security:** Standards embed secure practices into systems
- **Global consistency:** The Internet works the same worldwide

Standards of Networking

Major Organizations that create Networking Standards

1. IEEE (Institute of Electrical and Electronics Engineers)

- IEEE develops many networking standards, especially for physical and data link layers.
- **Examples:**
 - IEEE 802.3 — Ethernet
 - IEEE 802.11 — Wi-Fi
 - IEEE 802.15 — Bluetooth and PAN technologies

2. IETF (Internet Engineering Task Force)

- Responsible for Internet standards, especially TCP/IP protocols. Publishes **RFCs (Requests for Comments)**, many of which become Internet standards.
- **Examples:**
 - TCP, UDP, IP protocols
 - DNS
 - HTTP, SMTP, FTP rules

Standards of Networking

Major Organizations that create Networking Standards

3. ISO (International Organization for Standardization)

- Creates international standards, including the famous **OSI Model**, which guides network design.
- **Example:**
 - ISO/IEC 7498 — OSI Reference Model

4. ITU-T (International Telecommunication Union – Telecommunication Standardization Sector)

- Focuses on telecommunications and global communication standards.
- **Examples:**
 - Telephone systems
 - Broadband standards
 - Video conferencing protocols (e.g., H.323)

5. W3C (World Wide Web Consortium)

- Develops standards for the World Wide Web.
- **Examples:**
 - HTML, CSS
 - Web accessibility
 - Web architecture guidelines

Standards of Networking

Examples of important Networking Standards

- **Ethernet (IEEE 802.3):** Defines wired LAN technology
- **Wi-Fi (IEEE 802.11):** Defines wireless LAN communication
- **IPv4 and IPv6:** Internet addressing protocols
- **HTTPS:** Secure web communication
- **USB standards:** For device connectivity
- **5G/4G standards:** Mobile communication

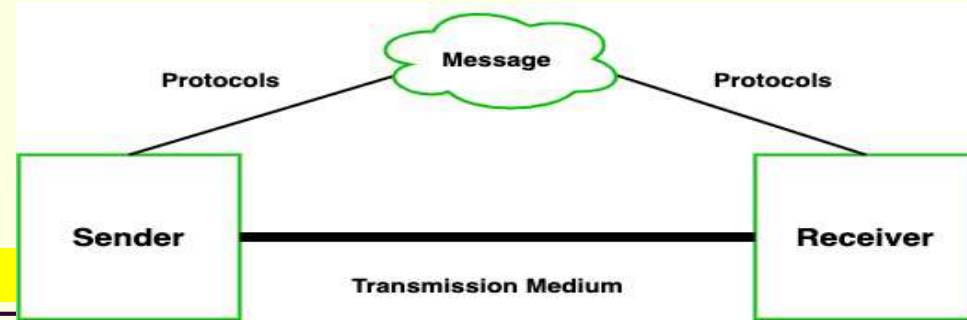
How Standards Improve Networking

- Reduce equipment cost through uniformity
- Allow innovation by providing a common framework
- Enable large-scale systems like cloud computing and the global Internet
- Improve security and reliability

Data Communication – Definition, Components, Types, Channels

- Definition
- Communication is defined as a process in which more than one computer transfers information, instructions to each other and for sharing resources.
- Or in other words, communication is a process or act in which we can send or receive data.
- A network of computers is defined as an interconnected collection of autonomous computers. Autonomous means no computer can start, stop or control another computer.

Data Communication



- **Components of Data Communication**
- A communication system is made up of the following components:
- **Message:** A message is a piece of information that is to be transmitted from one person to another. It could be a text file, an audio file, a video file, etc.
- **Sender:** It is simply a device that sends data messages. It can be a computer, mobile, telephone, laptop, video camera, or workstation, etc.
- **Receiver:** It is a device that receives messages. It can be a computer, telephone mobile, workstation, etc.
- **Transmission Medium / Communication Channels:** Communication channels are the medium that connect two or more workstations. Workstations can be connected by either wired media or wireless media.
- **Set of rules (Protocol):** When someone sends the data (The sender), it should be understandable to the receiver also otherwise it is meaningless. **For example, Sonali sends a message to Chetan. If Sonali writes in Hindi and Chetan cannot understand Hindi, it is a meaningless conversation.**

COMPONENTS OF A COMPUTER NETWORK

1. Nodes
2. Media
3. Services *

1. NODES

End nodes (End devices).

Intermediary nodes.

END NODES(DEVICES)

Computers

Network Printers

VoIP Phones

Telepresence endpoint

Security cameras

Mobile handheld devices (Smart phones, tablets, PDAs, Wireless debit/credit card reader, barcode scanner)

INTERMEDIARY NODES

Switches

Wireless Access Point

Routers

Security Devices (Firewall)

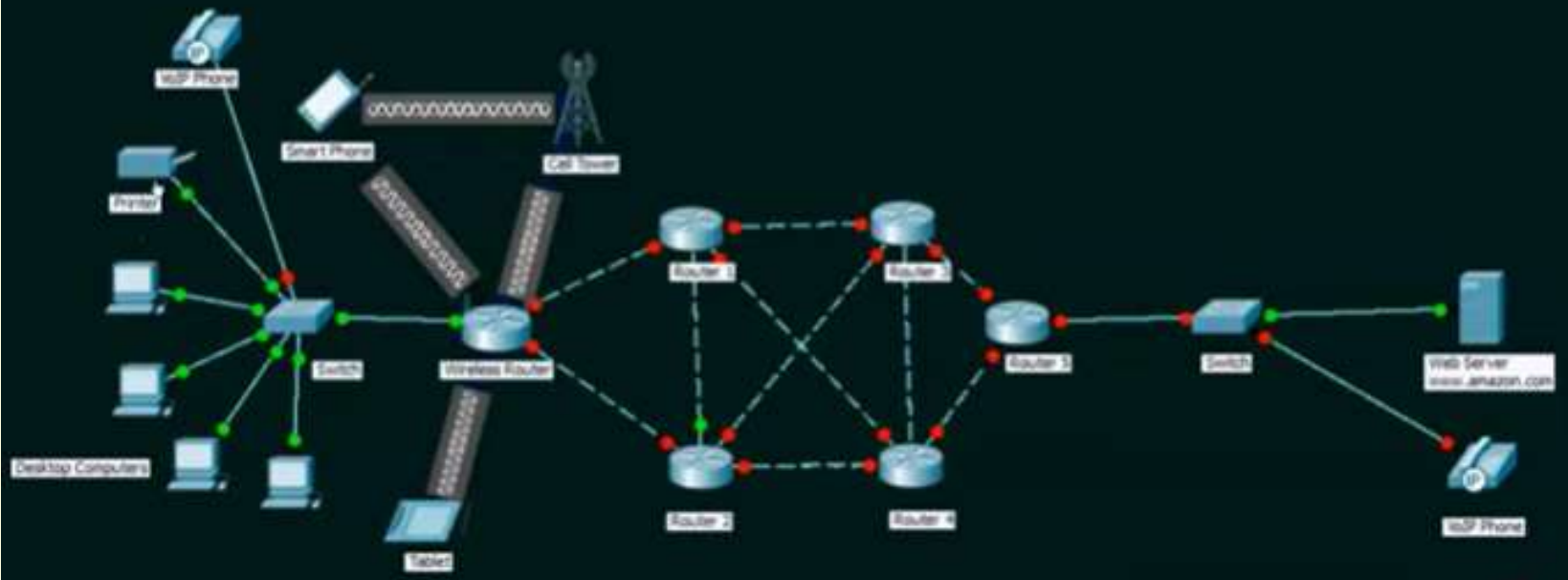
Bridges

Hubs

Repeaters

Cell Tower

COMPONENTS – NODES, MEDIA AND SERVICES



2. MEDIA

Wired Medium (Guided Medium)

Wireless Medium (Unguided Medium)

WIRED MEDIA

Ethernet straight-through cable

Ethernet crossover cable

Fiber Optic cable

Coaxial cable

USB cable

WIRED MEDIA

Ethernet straight-through cable

Ethernet crossover cable

Fiber Optic cable

Coaxial cable

USB cable



WIRED MEDIA

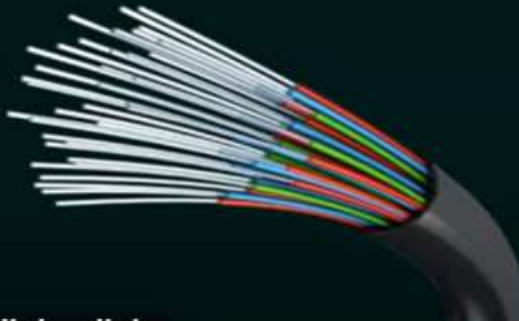
Ethernet straight-through cable

Ethernet crossover cable

Fiber Optic cable

Coaxial cable

USB cable



WIRED MEDIA

Ethernet straight-through cable

Ethernet crossover cable

Fiber Optic cable

Coaxial cable

USB cable



WIRED MEDIA

Ethernet straight-through cable

Ethernet crossover cable

Fiber Optic cable

Coaxial cable

USB cable



WIRELESS MEDIA

Infrared (Example: short range communication – TV remote control)

Radio (Example: Bluetooth, Wi-Fi)

Microwaves (Example: Cellular System)

Satellite (Example: Long range communication – GPS)

3. SERVICES

e-Mail

Storage services

File sharing

Instant messaging

Online game

Voice over IP

Video telephony

World Wide Web

PROTOCOLS

Protocol = Rule.

It is a set of rules that govern data communication.

Protocol determines:

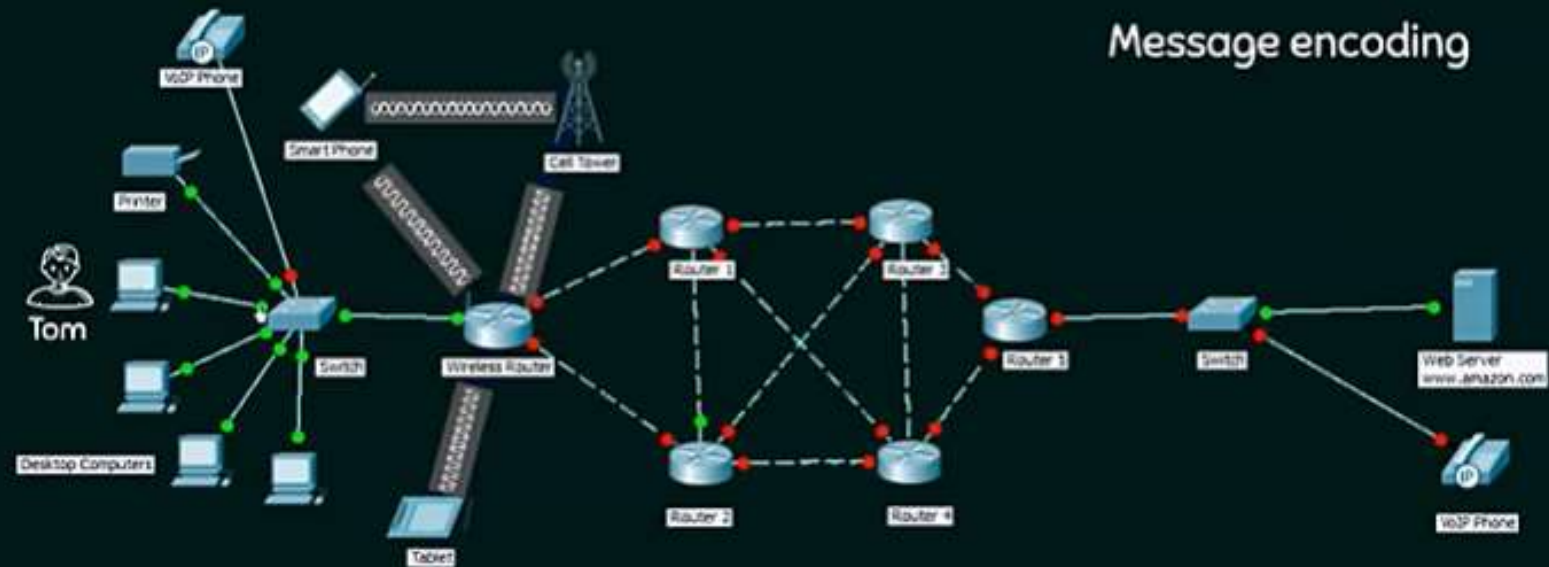
- ☐ What is communicated?
- ☐ How it is communicated?
- ☐ When it is communicated?

ELEMENTS OF A PROTOCOL

1. Message encoding
2. Message formatting and encapsulation
3. Message timing
4. Message size
5. Message delivery options

ELEMENTS OF PROTOCOL IN COMPUTER NETWORK

Message encoding



CLIENT SERVER NETWORK

- ★ Centralized administration.
- ★ Request-Response model.
- ★ Scalable.
- ★ Server may be overloaded.



PEER-TO-PEER NETWORK

- ★ No Centralized administration.
- ★ All peers are equal.
- ★ Simple sharing applications.
- ★ Not scalable.



Data Communication

- Types of data communication
- **Simplex Communication:** It is one-way communication or we can say that unidirectional communication in which one device only receives and another device only sends data and devices use their entire capacity in transmission. For example, IoT, entering data using a keyboard, listening music using a speaker, etc.
- **Half Duplex communication:** It is a two-way communication, or we can say that it is a bidirectional communication in which both the devices can send and receive data but not at the same time. When one device is sending data then another device is only receiving and vice-versa. For example, walkie-talkie.
- **Full-duplex communication:** It is a two-way communication or we can say that it is a bidirectional communication in which both the devices can send and receive data at the same time. For example, mobile phones, landlines, etc.

Data Communication

- Communication Channels
- Communication channels are the medium that connects two or more workstations. Workstations can be connected by either wired media or wireless media. It is also known as a transmission medium. The transmission medium or channel is a link that carries messages between two or more devices.
- We can group the communication media into two categories:
 - Guided media transmission
 - Unguided media transmission

Data Communication

- Communication Media
- There are 3 major types of Guided Media: Twisted pair cable, Coaxial cable, Optical fiber cable
- Twisted Pair Cable
- It consists of 2 separately insulated conductor wires wound about each other. Generally, several such pairs are bundled together in a protective sheath. They are the most widely used Transmission Media.
- Twisted Pair is of two types:
 - Unshielded Twisted Pair (UTP)
 - Shielded Twisted Pair (STP)

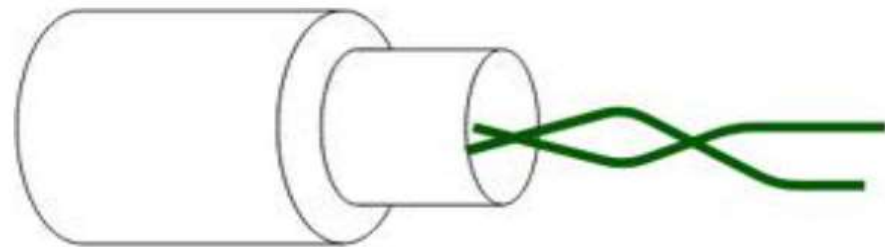
Data Communication

- Twisted Pair is of two types:

- **Unshielded Twisted Pair (UTP):** UTP consists of two insulated copper wires twisted around one another. This type of cable has the ability to block interference and does not depend on a physical shield for this purpose. It is used for telephonic applications.
- **Shielded Twisted Pair (STP):** This type of cable consists of a special jacket (a copper braid covering or a foil shield) to block external interference. It is used in fast-data-rate Ethernet and in voice and data channels of telephone lines.



Unshielded Twisted Pair



Shielded Twisted Pair

Data Communication

- Coaxial Cable
- It has an outer plastic covering containing an insulation layer made of PVC or Teflon and 2 parallel conductors each having a separate insulated protection cover. The coaxial cable transmits information in two modes: Baseband mode (dedicated cable bandwidth) and Broadband mode (cable bandwidth is split into separate ranges). Cable TVs and analog television networks widely use Coaxial cables.

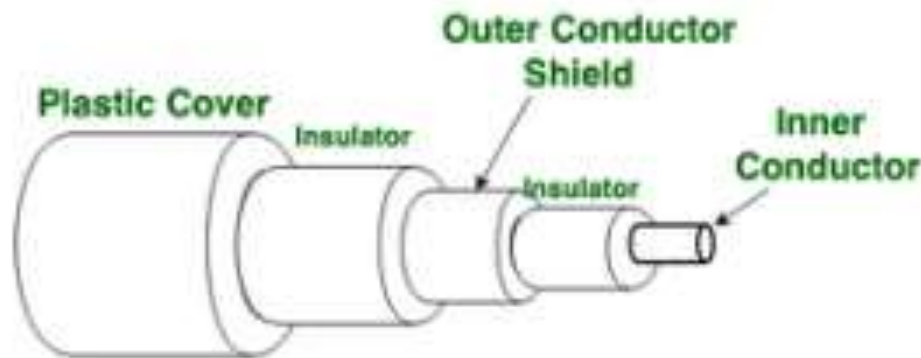
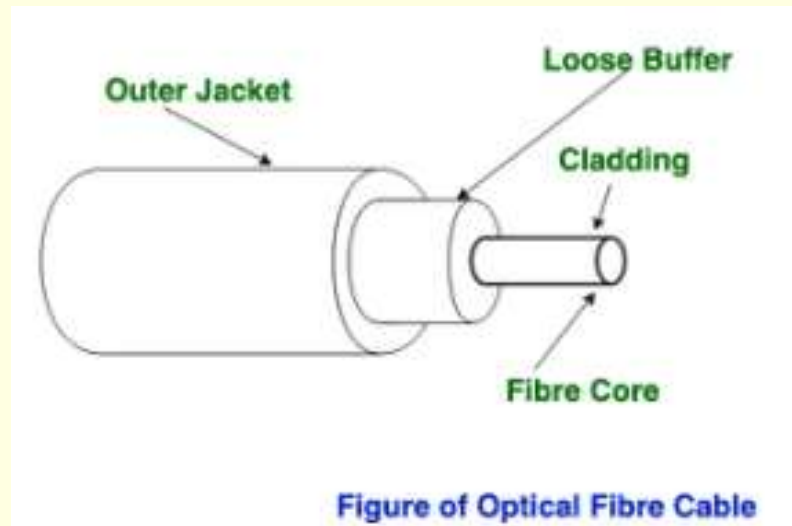


Figure of Coaxial Cable

Data Communication

- Optical Fiber Cable
- Optical Fibre Cable uses the concept of refraction of light through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called the cladding. It is used for the transmission of large volumes of data. The cable can be unidirectional or bidirectional. The WDM (Wavelength Division Multiplexer) supports two modes, namely unidirectional and bidirectional mode.



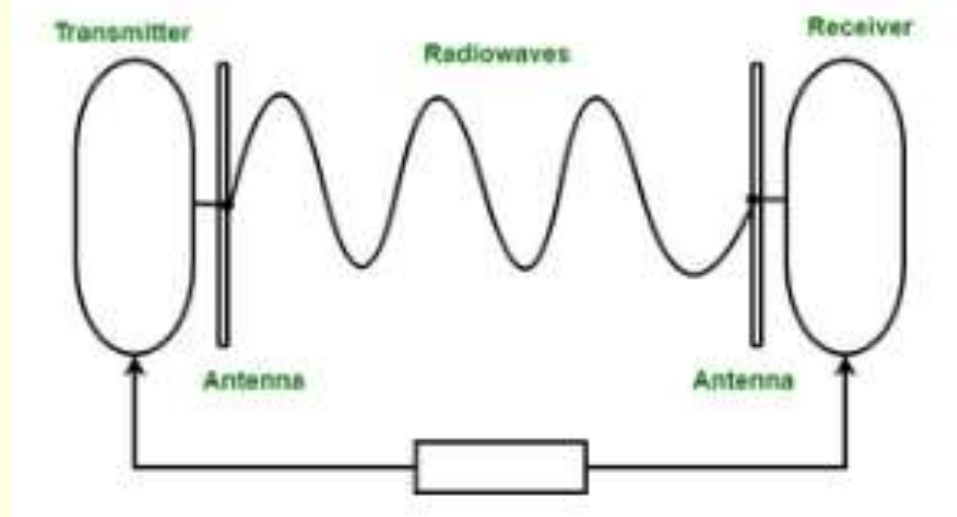
Data Communication

- Communication Media

- There are 3 major types of Unguided Media: Radio waves, Microwaves, Infrared

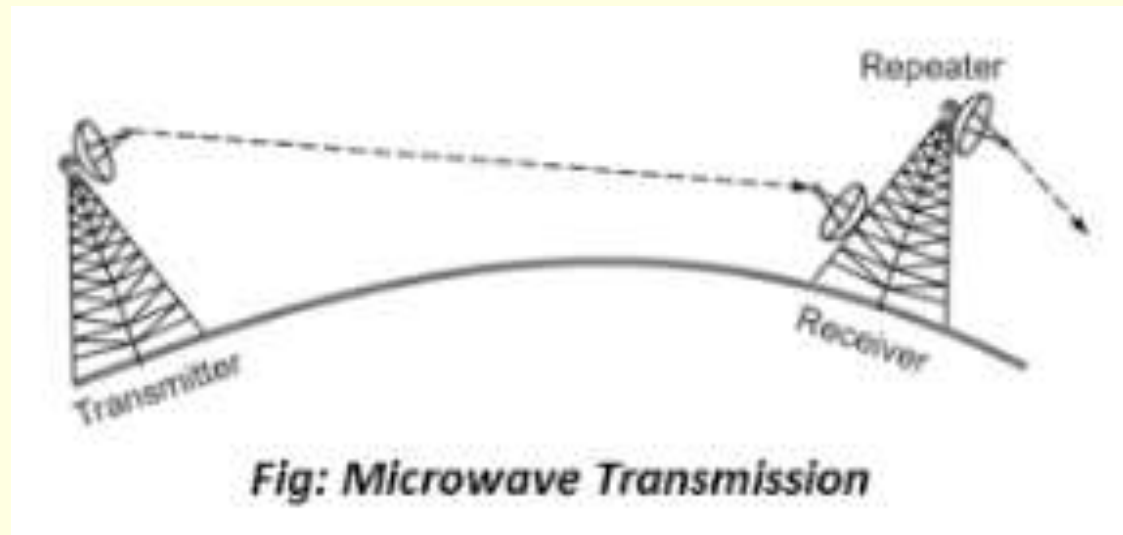
- Radio Waves

- Radio waves are easy to generate and can penetrate through buildings. The sending and receiving antennas need not be aligned. Frequency Range: 3KHz – 1GHz. AM and FM radios and cordless phones use Radio waves for transmission.



Data Communication

- **Microwaves**
- It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other. The distance covered by the signal is directly proportional to the height of the antenna. **Frequency Range: 1GHz – 300GHz.** Micro waves are majorly used for mobile phone communication and television distribution.



Data Communication

- Infrared
- Infrared waves are used for very short distance communication. They cannot penetrate through obstacles. This prevents interference between systems.
Frequency Range: 300GHz – 400THz. It is used in TV remotes, wireless mouse, keyboard, printer, etc.

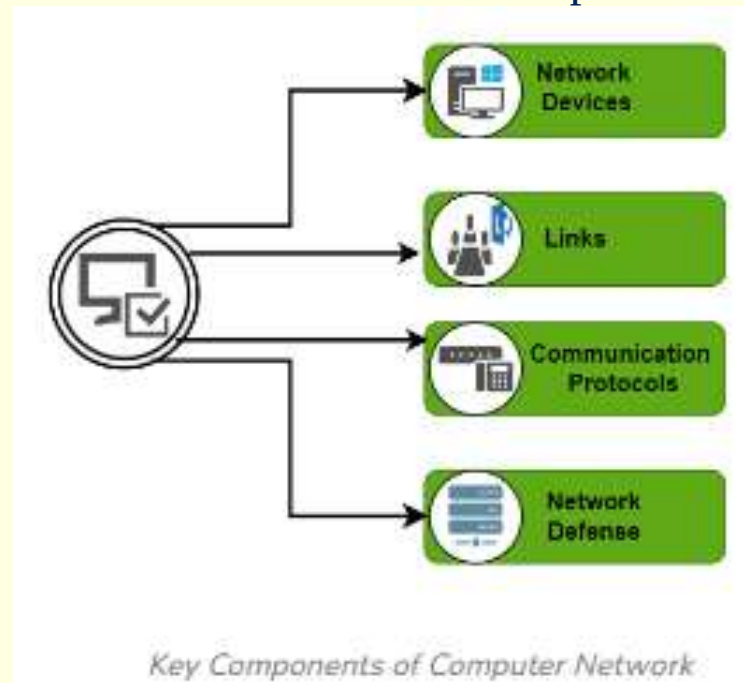


Computer Network

- **Computer Network:** A computer network is a system that connects many independent computers to share information (data) and resources. The integration of computers and other different devices allows users to communicate more easily.
- A computer network is a collection of two or more computer systems that are linked together. A network connection can be established using either cable or wireless media. Hardware and software are used to connect computers and tools in any network.

Key components of Computer Network

- **Key components of a Computer Network:** In simple terms, a computer network is made up of two main parts: devices (called nodes) and connections (called links). The links connect the devices to each other. The rules for how these connections send information are called communication protocols. The starting and ending points of these communications are often called ports.



Network Devices



1. Network Devices

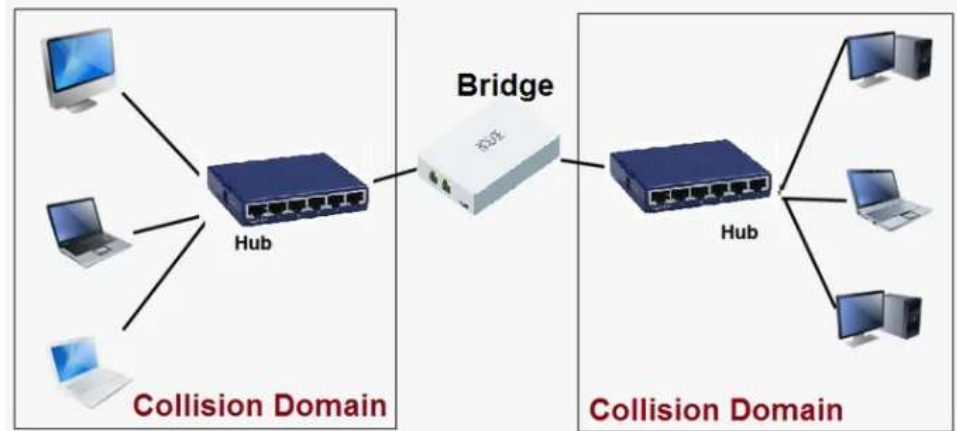
- Basic hardware interconnecting network nodes, such as Network Interface Cards (NICs), Bridges, Hubs, Switches, and Routers, are used in all networks. In addition, a mechanism for connecting these building parts is necessary, which is usually galvanic cable and optical cable are less popular (“optical fiber”). The following are the network devices:
- **NIC (Network Interface Card):** A network card, often known as a network adapter or NIC (network interface card), is computer hardware that enables computers to communicate via a network. It offers physical access to networking media and, in many cases, MAC addresses serve as a low-level addressing scheme. Each network interface card has a distinct identifier. This is stored on a chip that is attached to the card.

Network Devices



- **Repeater:** A repeater is an electrical device that receives a signal, cleans it of unwanted noise, regenerates it, and retransmits it at a higher power level or to the opposite side of an obstruction, allowing the signal to travel greater distances without degradation. In the majority of twisted pair Ethernet networks, Repeaters are necessary for cable lengths longer than 100 meters in some systems. Repeaters are based on physics.
- **Hub:** A hub is a device that joins together many twisted pairs or fiber optic Ethernet devices to give the illusion of a formation of a single network segment. The device can be visualized as a multiport repeater. A network hub is a relatively simple broadcast device. Any packet entering any port is regenerated and broadcast out on all other ports, and hubs do not control any of the traffic that passes through them. Packet collisions occur as a result of every packet being sent out through all other ports, substantially impeding the smooth flow of communication.

Network Devices



- **Bridges:** Bridges broadcast data to all the ports but not to the one that received the transmission. Bridges, on the other hand, learn which MAC addresses are reachable through specific ports rather than copying messages to all ports as hubs do. Once a port and an address are associated, the bridge will only transport traffic from that address to that port.
- **Switches:** A switch differs from a hub in that it only forwards frames to the ports that are participating in the communication, rather than all of the ports that are connected. The collision domain is broken by a switch, yet the switch depicts itself as a broadcast domain. Frame-forwarding decisions are made by switches based on MAC addresses.

-
- A **Collision Domain** is a scenario in which when a device sends out a message to the network, all other devices which are included in its collision domain have to pay attention to it, no matter if it was destined for them or not. This causes a problem because, in a situation where two devices send out their messages simultaneously, a collision will occur leading them to wait and re-transmit their respective messages, one at a time. Remember, it happens only in the case of a half-duplex mode.
 - A **Broadcast Domain** is a scenario in which when a device sends out a broadcast message, all the devices present in its broadcast domain have to pay attention to it. This creates a lot of congestion in the network, commonly called LAN congestion, which affects the bandwidth of the users present in that network.
 - From this, we can realize that the more the number of collision domains and the more the number of broadcast domains, the more efficient is the network providing better bandwidth to all its users.

Advantages of Collision Domain:

- **High Network Performance:** Collision Domain helps to improve network performance by reducing collisions on the network, which can improve data transmission and reduce packet loss.

Disadvantages of Collision Domain:

- **Limited Scalability:** Collision Domain may not be scalable in larger networks, as the number of devices connected to the network increases, which can lead to network congestion and performance degradation.

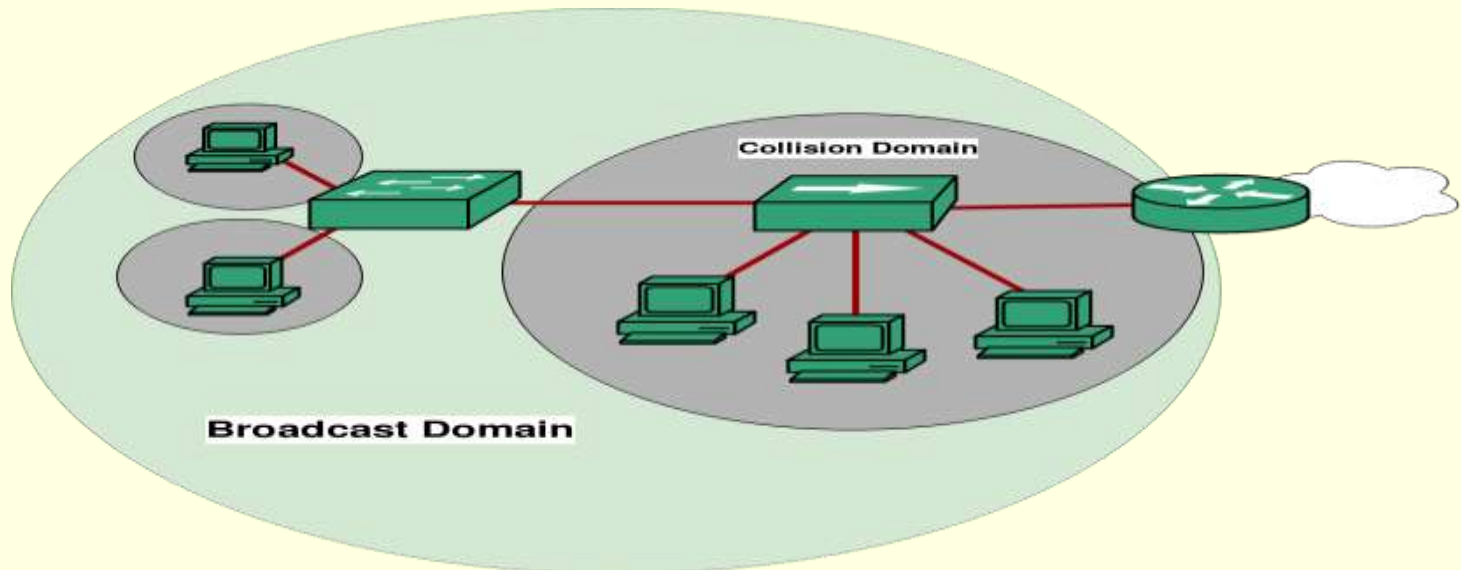
Advantages of Broadcast Domain:

- **Efficient Network Communication:** Broadcast Domain enables efficient network communication by allowing multiple devices to receive the same message simultaneously.

Disadvantages of Broadcast Domain:

- **Increased Network Congestion:** Broadcast Domain can lead to increased network congestion, particularly in larger networks, which can impact network performance and lead to packet loss.

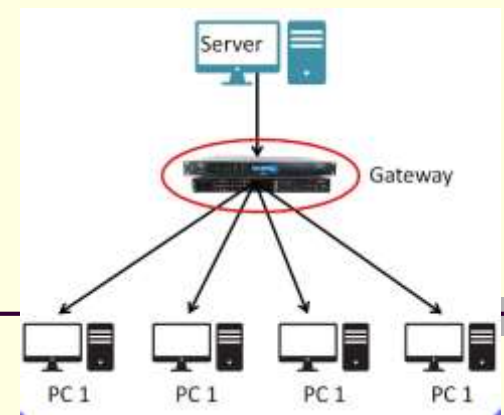
- A **router** not only breaks collision domains but also breaks broadcast domains, which means it is both collisions as well as broadcast domain separators. A router creates a connection between two networks. A broadcast message from one network will never reach the other one as the router will never let it pass.



- Also, as repeaters and bridges differ from hubs and switches only in terms of the number of ports, a **repeater** does not break collision and broadcast domains, while a **bridge** breaks only collision domains.



Network Devices



- **Routers:** Routers are networking devices that use headers and forwarding tables to find the optimal way to forward data packets between networks. A router is a computer networking device that links two or more computer networks and selectively exchanges data packets between them. A router can use address information in each data packet to determine if the source and destination are on the same network or if the data packet has to be transported between networks.
- Gateways:** To provide system compatibility, a gateway may contain devices such as protocol translators, impedance-matching devices, rate converters, fault isolators, or signal translators. It also necessitates the development of administrative procedures that are acceptable to both networks. By completing the necessary protocol conversions, a protocol translation / mapping gateway joins networks that use distinct network protocol technologies.

Key components of computer network – network devices



Modem



NIC



Repeater



Hub



Switch



Router



Bridge



Gateway

Key components of computer network – links

2. Links

- Links are the ways information travels between devices, and they can be of two types:
- **Wired:** Communication done in a wired medium. Copper wire, twisted pair, or fiber optic cables are all options. A wired network employs wires to link devices to the Internet or another network, such as laptops or desktop PCs.
- **Wireless:** Wireless means without wire, media that is made up of electromagnetic waves (EM Waves) or infrared waves. Antennas or sensors will be present on all wireless devices. For data or voice communication, a wireless network uses radio frequency waves rather than wires.

Key components of computer network – communication protocols

3. Communication Protocols

- A communication protocol is a set of rules that all devices follow when they share information. Some common protocols are TCP/IP, IEEE 802, Ethernet, wireless LAN, and cellular standards. TCP/IP is a model that organizes how communication works in modern networks. It has four functional layers for these communication links:
- **Network Access Layer:** This layer controls how data is physically transferred, including how hardware sends data through wires or fibers.
- **Internet Layer:** This layer packages data into understandable packets and ensures it can be sent and received.
- **Transport Layer:** This layer keeps the communication between devices steady and reliable.
- **Application Layer:** This layer allows high-level applications to access the network to start data transfer.

Key components of computer network – communication protocols

- Most of the modern internet structure is based on the TCP/IP model, although the similar seven-layer OSI model still has a strong influence.
- IEEE 802 is a group of standards for local area networks (LAN) and metropolitan area networks (MAN). The most well-known member of the IEEE 802 family is wireless LAN, commonly known as WLAN or Wi-Fi.

Key components of computer network – network defense

4. Network Defense

- While nodes, links, and protocols are the building blocks of a network, a modern network also needs strong defenses.
- Security is crucial because huge amounts of data are constantly being created, moved, and processed.
- Some examples of network defense tools are firewalls, intrusion detection systems (IDS), intrusion prevention systems (IPS), network access control (NAC), content filters, proxy servers, anti-DDoS devices, and load balancers.

Criteria of a good network

- **Criteria of a Good Network**
- **Performance:** It can be measured in many ways, including transmit time and response time. Transit time is the amount of time required for a message to travel from one device to another. Response time is the elapsed time between an inquiry and a response. The performance of the network depends on a number of factors, including the number of users, the type of medium & Hardware.
- **Reliability:** In addition to accuracy is measured by frequency of failure, the time it takes a link to recover from failure, and the network's robustness in catastrophe.
- **Security:** Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data loss.

Types of computer network

- Types of Computer Networks - Division Based on Area Covered
- Local Area Network (LAN): A LAN is a network that covers an area of around 10 kilometers. For example, a college network or an office network. Depending upon the needs of the organization, a LAN can be a single office, building, or Campus. Each host in LAN has an identifier, an address that defines hosts in LAN. A packet sent by the host to another host carries both the source host's and the destination host's address.
- Metropolitan Area Network (MAN): MAN refers to a network that covers an entire city. For example: consider the cable television network.
- Wide Area Network (WAN): WAN refers to a network that connects countries or continents. For example, the Internet allows users to access a distributed system called www from anywhere around the globe. WAN interconnects connecting devices such as switches, routers, or modems.

Types of computer network

- Types of Computer Networks - Division Based on Types of Communication
- Point To Point networks: Point-to-Point networking is a type of data networking that establishes a direct link between two networking nodes. A direct link between two devices, such as a computer and a printer, is known as a point-to-point connection.
- Multipoint: is the one in which more than two specific devices share links. In the multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is a spatially shared connection.
- Broadcast networks: In broadcast networks, a signal method in which numerous parties can hear a single sender. Radio stations are an excellent illustration of the “Broadcast Network” in everyday life. The radio station is a sender of data / signal in this scenario, and data is only intended to travel in one direction. Away from the radio transmission tower, to be precise.

Types of computer network

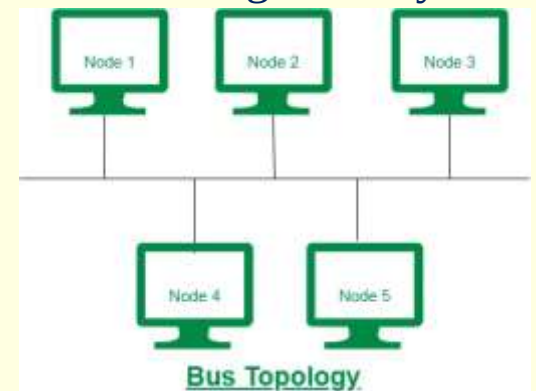
- Types of Computer Networks - Division Based on the Type of Architecture
- P2P Networks: Computers with similar capabilities and configurations are referred to as peers. The “peers” in a peer-to-peer network are computer systems that are connected to each other over the Internet. Without the use of a central server, files can be shared directly between systems on the network.
- Client-Server Networks: Each computer or process on the network is either a client or a server in a client-server architecture (client/server). The client asks for services from the server, which the server provides. Servers are high-performance computers or processes that manage disc drives (file servers), printers (print servers), or network traffic (network servers)
- Hybrid Networks: The hybrid model uses a combination of client-server and peer-to-peer architecture. For example: Torrent.

Network Topology

- What is Network Topology?
- The structure of the network and how each component is connected to the others are defined by the network topology. Different types of network topology are mentioned below:
 - Bus Topology
 - Ring Topology
 - Star Topology
 - Mesh Topology
 - Tree Topology

Network Topology

- **Bus Topology**
- Every computer and network device is connected to a single cable in a bus topology network. Linear Bus topology is defined as having exactly two terminals.



Advantages

- Installation is simple
- Compared to mesh, star, and tree topologies, the bus utilizes less cabling

Disadvantages

- Difficulty in reconfiguring and isolating faults
- A bus cable malfunction or break interrupts all communication

Network Topology

- Ring Topology

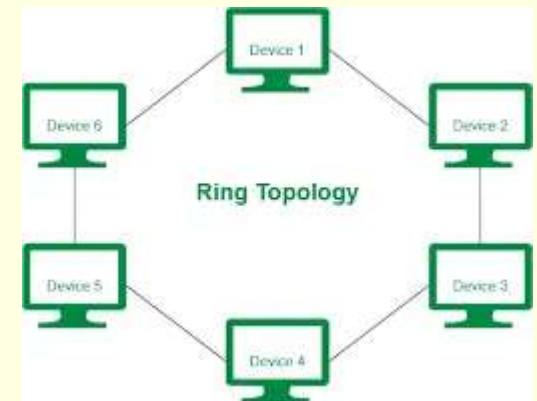
- The topology is named ring topology because one computer is connected to another, with the final one being connected to the first. Exactly two neighbors for each device. A signal is passed along the ring in one direction. Each ring incorporates a repeater.

Advantages

- Data transmission is relatively straightforward because packets only move in one direction
- There is no requirement for a central controller to manage communication between nodes
- Easy installation & Reconfiguration and Simplified Faulty connections

Disadvantages

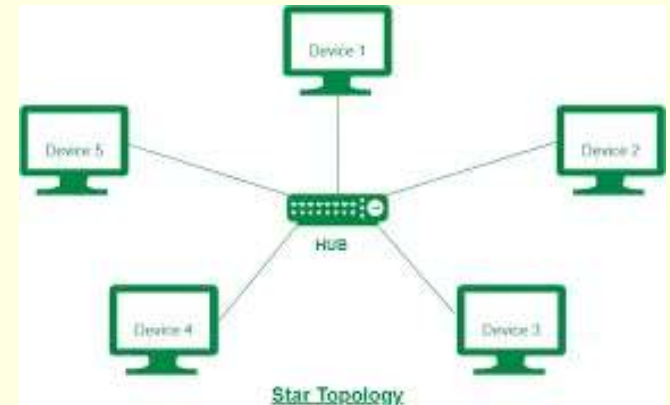
- In a Unidirectional Ring, a data packet must traverse through all nodes
- All computers must be turned on in order for them to connect with one another



Network Topology

■ Star Topology

- Each device in a star topology has a dedicated point-to-point link to a central controller, which is commonly referred to as the HUB. There is no direct connection between the devices. Traffic between the devices is not allowed in this topology. As an exchange, the controller is used.



Advantages

- When attaching or disconnecting devices, there are no network interruptions
- It's simple to set up and configure; Identifying and isolating faults is simple
- Less Expensive than mesh and Easy to install & configure

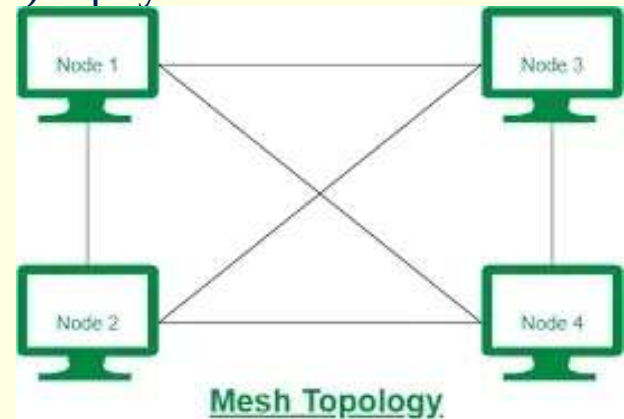
Disadvantages

- Nodes attached to the hub, switch, or concentrator is failed if they fail
- Because of the expense of the hubs, it is more expensive than linear bus topologies
- More cable is required compared to a bus or ring; Too much dependency on Hub

Network Topology

■ Mesh Topology

- Every device in a mesh topology has dedicated point-to-point connectivity to every other device. The term “dedicated” refers to the fact that the link exclusively transports data between the two devices it links. To connect n devices, a fully connected mesh network contains $n*(n-1)/2$ physical channels.



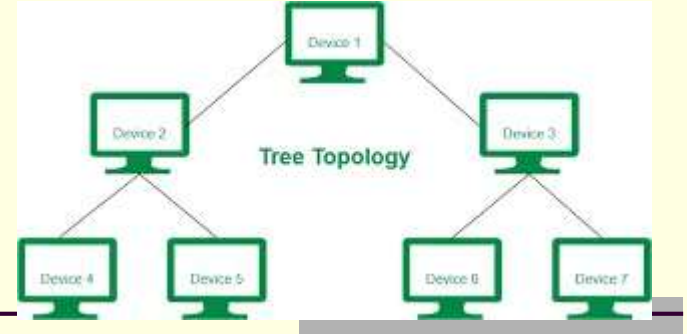
Advantages

- This topology can handle a lot of traffic.
- Even if one of the connections fails, a backup is always available. As a result, data transit is unaffected.
- Physical boundaries prevent other users from gaining access to messages.
- Point to Point links make fault transmission & fault isolation easy.

Disadvantages

- The amount of cabling and the number of I/O ports that are necessary.
- It is difficult to install and reconfigure.

Network Topology



Tree Topology

- The topology of a tree is similar to that of a star. Nodes in a tree, like those in a star, are connected to a central hub that manages network traffic. It has a root node, which is connected to all other nodes, producing a hierarchy. Hierarchical topology is another name for it. The number of Star networks is connected via Bus in Tree Topology.

Advantages

- Network expansion is both possible and simple.
- We partition the entire network into pieces (star networks) that are easier to manage and maintain.
- Other segments are unaffected if one segment is damaged.

Disadvantages

- Tree topology relies largely on the main bus cable because of its basic structure, and if it fails, the entire network is handicapped.
- Maintenance becomes more challenging when more nodes and segments are added.

Reference Models / Network Architecture

Reference Models (Network Architectures):

- OSI reference model
- TCP / IP reference Model

Reference Models / Network Architecture

- OSI stands for Open Systems Interconnection. The OSI model, created in 1984 by ISO, is a reference framework that explains the process of transmitting data between computers. It is divided into seven layers that work together to carry out specialised network functions, allowing for a more systematic approach to networking.



Reference Models / Network Architecture

- Physical Layer – Layer 1
- The lowest layer of the OSI reference model is the physical layer. It is responsible for the actual physical connection between the devices. The physical layer contains information in the form of bits. It is responsible for transmitting individual bits from one node to the next. When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together.

Reference Models / Network Architecture

Functions of the Physical Layer

- **Bit Synchronization:** The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at the bit level.
- **Bit Rate Control:** The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- **Physical Topologies:** Physical layer specifies how the different, devices/nodes are arranged in a network i.e. bus, star, or mesh topology.
- **Transmission Mode:** Physical layer also defines how the data flows between the two connected devices. The various transmission modes possible are Simplex, half-duplex and full-duplex.

Reference Models / Network Architecture

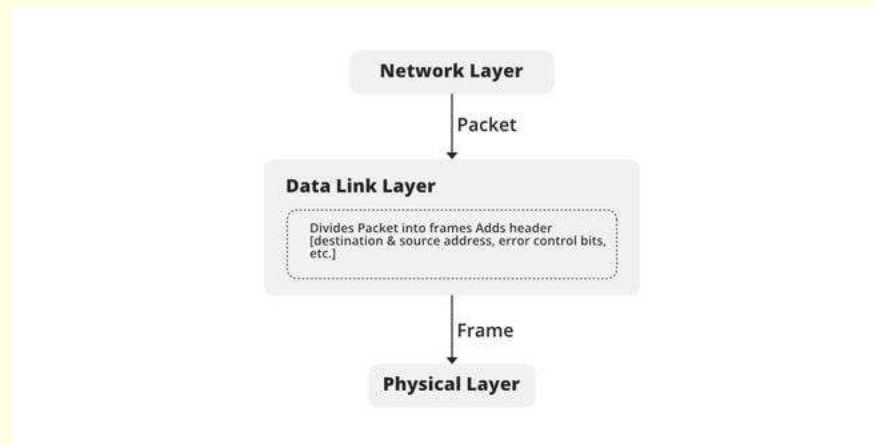
- Data Link Layer (DLL) – Layer 2
- The data link layer is responsible for the node-to-node delivery of the message. The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer. When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its MAC address.

Reference Models / Network Architecture

- **Functions of the Data Link Layer**
- **Framing:** Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.
- **Physical Addressing:** After creating frames, the Data link layer adds physical addresses (MAC addresses) of the sender and/or receiver in the header of each frame.
- **Error Control:** The data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.
- **Flow Control:** The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- **Access Control:** When a single communication channel is shared by multiple devices, the MAC sub-layer of the data link layer helps to determine which device has control over the channel at a given time.

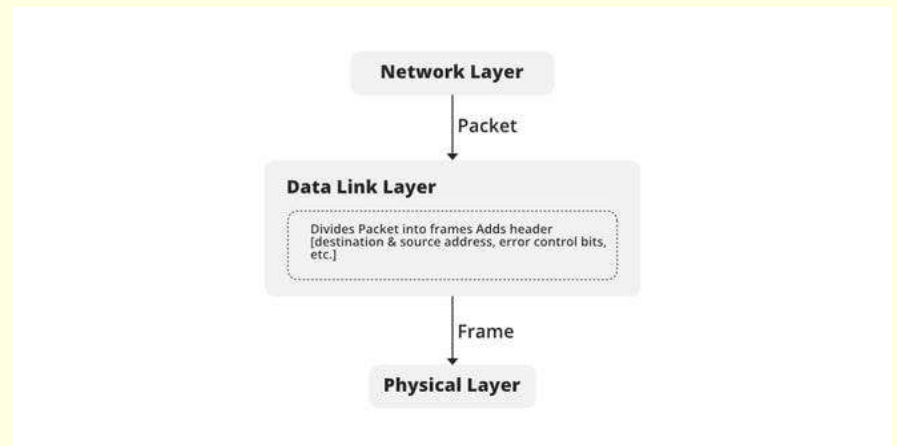
Reference Models / Network Architecture

- Network Layer – Layer 3
- The network layer works for the transmission of data from one host to the other located in different networks. It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available. The sender & receiver's IP addresses are placed in the header by the network layer.



Reference Models / Network Architecture

- **Functions of the Network Layer**
- **Routing:** The network layer protocols determine which route is suitable from source to destination. This function of the network layer is known as routing.
- **Logical Addressing:** To identify each device inter-network uniquely, the network layer defines an addressing scheme. The sender & receiver's IP addresses are placed in the header by the network layer. Such an address distinguishes each device uniquely and universally.



Reference Models / Network Architecture

- Transport Layer – Layer 4
- The transport layer provides services to the application layer and takes services from the network layer. The data in the transport layer is referred to as Segments. It is responsible for the end-to-end delivery of the complete message. The transport layer also provides the acknowledgment of the successful data transmission and re-transmits the data if an error is found.
- At the sender's side: The transport layer receives the formatted data from the upper layers, performs Segmentation, and also implements Flow and error control to ensure proper data transmission. It also adds Source and Destination port numbers in its header and forwards the segmented data to the Network Layer.
- At the receiver's side: Transport Layer reads the port number from its header and forwards the Data which it has received to the respective application. It also performs sequencing and reassembling of the segmented data.

Reference Models / Network Architecture

- **Functions of the Transport Layer**
- **Segmentation and Reassembly:** This layer accepts the message from the (session) layer, and breaks the message into smaller units. Each of the segments produced has a header associated with it. The transport layer at the destination station reassembles the message.
- **Service Point Addressing:** To deliver the message to the correct process, the transport layer header includes a type of address called service point address or port address. Thus by specifying this address, the transport layer makes sure that the message is delivered to the correct process.

Reference Models / Network Architecture

Services Provided by Transport Layer

- **1. Connection-Oriented Service:** It is a three-phase process that includes:
 - Connection Establishment
 - Data Transfer
 - Termination/disconnection

In this type of transmission, the receiving device sends an acknowledgment, back to the source after a packet or group of packets is received. This type of transmission is reliable and secure.

- **2. Connectionless service:** It is a one-phase process and includes Data Transfer. In this type of transmission, the receiver does not acknowledge receipt of a packet. This approach allows for much faster communication between devices.
- Connection-oriented service is more reliable than connectionless Service.**

Reference Models / Network Architecture

- Session Layer – Layer 5
- This layer is responsible for the establishment of connection, maintenance of sessions, and authentication, and also ensures security.
- Functions of the Session Layer
- Session Establishment, Maintenance, and Termination: The layer allows the two processes to establish, use, and terminate a connection.
- Synchronization: This layer allows a process to add checkpoints that are considered synchronization points in the data. These synchronization points help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely and data loss is avoided.
- Dialog Controller: The session layer allows two systems to start communication with each other in half-duplex or full-duplex.

Reference Models / Network Architecture

- Presentation Layer – Layer 6
- The presentation layer is also called the Translation layer. The data from the application layer is extracted here and manipulated as per the required format to transmit over the network.
- Functions of the Presentation Layer
- Translation: For example, ASCII to EBCDIC.
- Encryption/ Decryption: Data encryption translates the data into another form or code. The encrypted data is known as the ciphertext and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
- Compression: Reduces the number of bits that need to be transmitted on the network.

Reference Models / Network Architecture

- Application Layer – Layer 7
- At the very top of the OSI Reference Model stack of layers, we find the Application layer which is implemented by the network applications. These applications produce the data to be transferred over the network. This layer also serves as a window for the application services to access the network and for displaying the received information to the user.
- Example: Application – Browsers, Skype Messenger, etc.

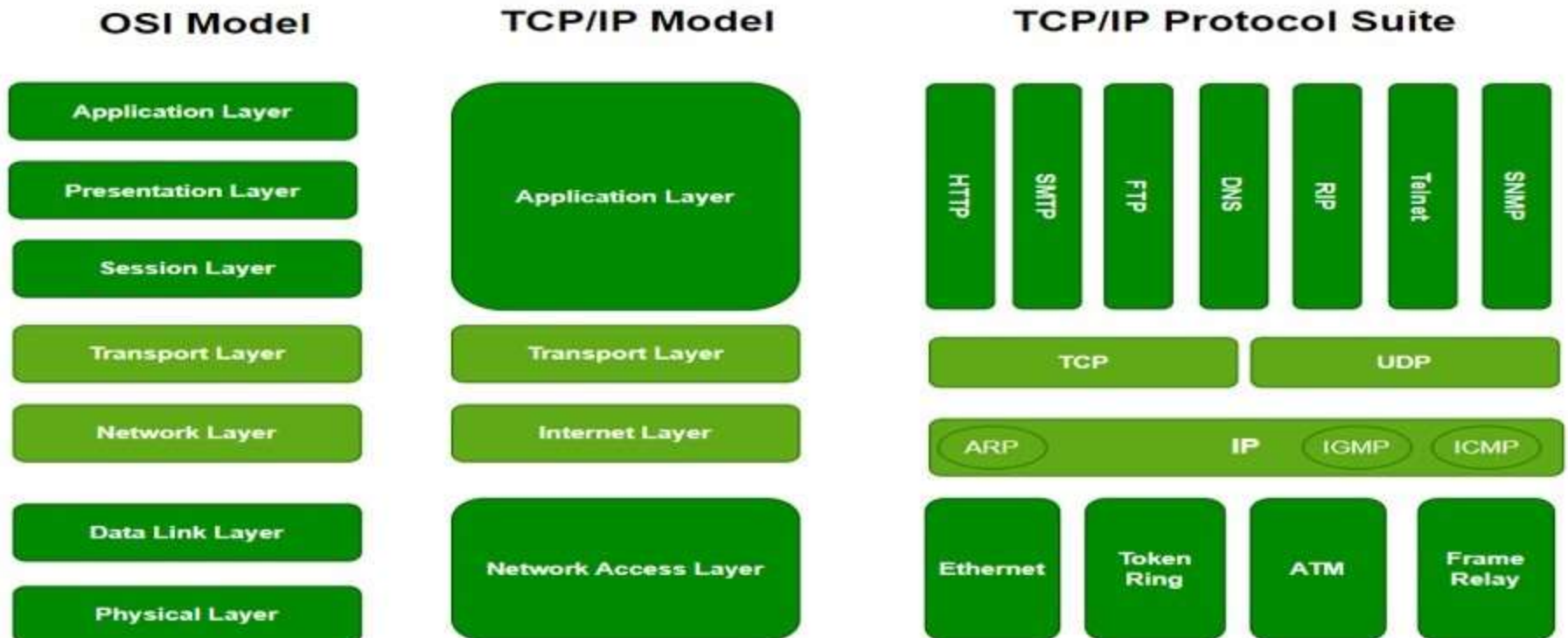
Reference Models / Network Architecture

- Functions of the Application Layer
- Network Virtual Terminal(NVT): It allows a user to log on to a remote host.
- File Transfer Access and Management(FTAM): This application allows a user to access files in a remote host, retrieve files in a remote host, and manage or control files from a remote computer.
- Mail Services: Provide email service.
- Directory Services: This application provides distributed database sources and access for global information about various objects and services.

Reference Models / Network Architecture

OSI vs TCP/IP Model

- TCP/IP protocol (Transfer Control Protocol/Internet Protocol) was created by U.S. Department of Defense's Advanced Research Projects Agency (ARPA) in 1970s.



Reference Models / Network Architecture

Some key differences between the OSI model and the TCP/IP Model are:

- TCP/IP model consists of 4 layers but OSI model has 7 layers. Layers 5,6,7 of the OSI model are combined into the Application Layer of TCP/IP model and OSI layers 1 and 2 are combined into Network Access Layers of TCP/IP protocol.
- The TCP/IP model is older than the OSI model, hence it is a foundational protocol that defines how should data be transferred online.
- Compared to the OSI model, the TCP/IP model has less strict layer boundaries.
- All layers of the TCP/IP model are needed for data transmission but in the OSI model, some applications can skip certain layers. Only layers 1,2 and 3 of the OSI model are necessary for data transmission.

Reference Models / Network Architecture

Advantages of OSI Model

- It divides network communication into 7 layers which makes it easier to understand and troubleshoot.
- It standardizes network communications, as each layer has fixed functions and protocols.
- Diagnosing network problems is easier with the OSI model.
- It is easier to improve with advancements as each layer can get updates separately.

Disadvantages of OSI Model

- **Complexity:** The OSI Model has seven layers, which can be complicated and hard to understand for beginners.
- **Not Practical:** In real-life networking, most systems use a simpler model called the Internet protocol suite (TCP/IP), so the OSI Model isn't always directly applicable.
- **Slow Adoption:** When it was introduced, the OSI Model was not quickly adopted by the industry, which preferred the simpler and already-established TCP/IP model.
- **Overhead:** Each layer in the OSI Model adds its own set of rules and operations, which can make the process more time-consuming and less efficient.
- **Theoretical:** The OSI Model is more of a theoretical framework, meaning it's great for understanding concepts but not always practical for implementation.

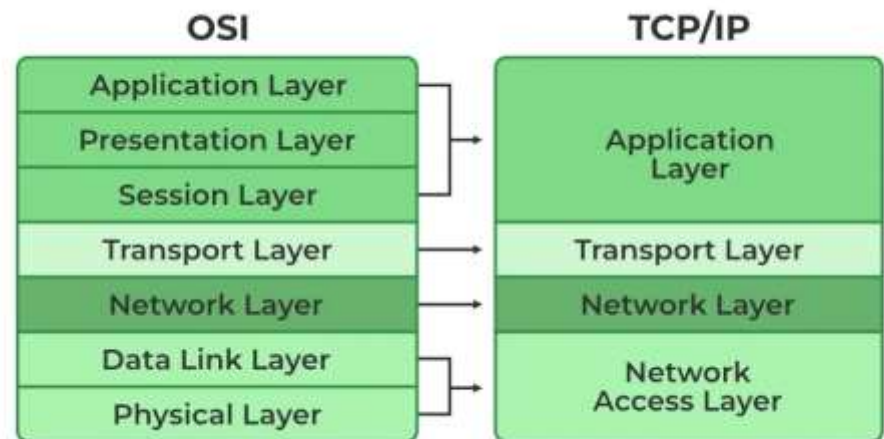
Reference Models / Network Architecture

■ TCP / IP reference Model

- The TCP/IP model is a fundamental framework for computer networking. It stands for Transmission Control Protocol/Internet Protocol, which are the core protocols of the Internet. This model defines how data is transmitted over networks, ensuring reliable communication between devices.

Layers of TCP/IP Model

- Application Layer
- Transport Layer(TCP/UDP)
- Network/Internet Layer(IP)
- Network Access Layer



Each layer has specific functions that help manage different aspects of network communication, making it essential for understanding and working with modern networks.

Reference Models / Network Architecture

- 1. Network Access Layer
- It is a group of applications requiring network communications. This layer is responsible for generating the data and requesting connections. It acts on behalf of the sender and the Network Access layer on the behalf of the receiver.
- The packet's network protocol type, in this case, TCP/IP, is identified by network access layer. Error prevention and “framing” are also provided by this layer. Point-to-Point Protocol (PPP) framing and Ethernet IEEE 802.2 framing are two examples of data-link layer protocols.

Reference Models / Network Architecture

- 2. Internet Layer
- This layer parallels the functions of OSI's Network layer. It defines the protocols which are responsible for the logical transmission of data over the entire network.
- The main protocols residing at Internet layer (IP, ICMP, ARP) are as follows:
- IP: IP stands for Internet Protocol and it is responsible for delivering packets from the source host to the destination host by looking at the IP addresses in the packet headers. IP has 2 versions: IPv4 and IPv6. IPv4 is the one that most websites are using currently. But IPv6 is growing as the number of IPv4 addresses is limited in number when compared to the number of users.

Reference Models / Network Architecture

- The main protocols residing at Internet layer are as follows:
- **ICMP:** ICMP stands for Internet Control Message Protocol. It is encapsulated within IP datagrams and is responsible for providing hosts with information about network problems.
- **ARP:** ARP stands for Address Resolution Protocol. Its job is to find the hardware address of a host from a known IP address. ARP has several types: Reverse ARP, Proxy ARP, Gratuitous ARP, and Inverse ARP.
- The Internet Layer is a layer in the Internet Protocol (IP) suite, which is the set of protocols that define the Internet. The Internet Layer is responsible for routing packets of data from one device to another across a network. It does this by assigning each device a unique IP address, which is used to identify the device and determine the route that packets should take to reach it.

Reference Models / Network Architecture

- 3. Transport Layer
- The TCP/IP transport layer protocols exchange data receipt acknowledgments and retransmit missing packets to ensure that packets arrive in order and without error. End-to-end communication is referred to as such. Transmission Control Protocol (TCP) and User Datagram Protocol are transport layer protocols at this level (UDP).

Reference Models / Network Architecture

- **TCP:** Applications can interact with one another using TCP as though they were physically connected by a circuit. TCP transmits data in a way that resembles character-by-character transmission rather than separate packets. A starting point that establishes the connection, the whole transmission in byte order, and an ending point that closes the connection make up this transmission.
- **UDP:** The datagram delivery service is provided by UDP, the other transport layer protocol. Connections between receiving and sending hosts are not verified by UDP. Applications that transport little amounts of data use UDP rather than TCP because it eliminates the processes of establishing and validating connections.

Reference Models / Network Architecture

- 4. Application Layer
- This layer is analogous to the transport layer of the OSI model. It is responsible for end-to-end communication and error-free delivery of data. It shields the upper-layer applications from the complexities of data.
- The three main protocols present in this layer are (HTTP / HTTPS, SSH, NTP) :
- HTTP and HTTPS: HTTP stands for Hypertext transfer protocol. It is used by the World Wide Web to manage communications between web browsers and servers. HTTPS stands for HTTP-Secure. It is a combination of HTTP with SSL(Secure Socket Layer). It is efficient in cases where the browser needs to fill out forms, sign in, authenticate, and carry out bank transactions.

Reference Models / Network Architecture

- The main protocols present in Application layer are:
- **SSH:** SSH stands for Secure Shell. It is a terminal emulations software similar to Telnet. The reason SSH is preferred is because of its ability to maintain the encrypted connection. It sets up a secure session over a TCP/IP connection.
- **NTP:** NTP stands for Network Time Protocol. It is used to synchronize the clocks on our computer to one standard time source. It is very useful in situations like bank transactions. Assume the following situation without the presence of NTP. Suppose you carry out a transaction, where your computer reads the time at 2:30 PM while the server records it at 2:28 PM. The server can crash very badly if it's out of sync.

Reference Models / Network Architecture

■ Difference between TCP/IP and OSI Model

TCP/IP	OSI
TCP refers to Transmission Control Protocol.	OSI refers to Open Systems Interconnection.
TCP/IP uses both the session and presentation layer in the application layer itself.	OSI uses different session and presentation layers.
TCP/IP follows connectionless a horizontal approach.	OSI follows a vertical approach.
The Transport layer in TCP/IP does not provide assurance delivery of packets.	In the OSI model, the transport layer provides assurance delivery of packets.
Protocols cannot be replaced easily in TCP/IP model.	While in the OSI model, Protocols are better covered and are easy to replace with the technology change.
TCP/IP model network layer only provides connectionless (IP) services. The transport layer (TCP) provides connections.	Connectionless and connection-oriented services are provided by the network layer in the OSI model.

Reference Models / Network Architecture

Advantages of TCP/IP Model

- **Interoperability:** The TCP/IP model allows different types of computers and networks to communicate with each other, promoting compatibility and cooperation among diverse systems.
- **Scalability:** TCP/IP is highly scalable, making it suitable for both small and large networks, from local area networks (LANs) to wide area networks (WANs) like the internet.
- **Standardization:** It is based on open standards and protocols, ensuring that different devices and software can work together without compatibility issues.
- **Flexibility:** The model supports various routing protocols, data types, and communication methods, making it adaptable to different networking needs.
- **Reliability:** TCP/IP includes error-checking and retransmission features that ensure reliable data transfer, even over long distances and through various network conditions.

Reference Models / Network Architecture

- **Disadvantages of TCP/IP Model**
- **Complex Configuration:** Setting up and managing a TCP/IP network can be complex, especially for large networks with many devices. This complexity can lead to configuration errors.
- **Security Concerns:** TCP/IP was not originally designed with security in mind. While there are now many security protocols available (such as SSL/TLS), they have been added on top of the basic TCP/IP model, which can lead to vulnerabilities.
- **Inefficiency for Small Networks:** For very small networks, the overhead and complexity of the TCP/IP model may be unnecessary and inefficient compared to simpler networking protocols.
- **Limited by Address Space:** Although IPv6 addresses this issue, the older IPv4 system has a limited address space, which can lead to issues with address exhaustion in larger networks.
- **Data Overhead:** TCP, the transport protocol, includes a significant amount of overhead to ensure reliable transmission. This can reduce efficiency, especially for small data packets or in networks where speed is crucial.

Computer Networks and Data Communication

